

Can You Spot the Error?

From Meiya:

This works:

[StandardCost] <>3

...but this does not work:

[StandardCost] <> 3

Expression Editor for Valid If



Data Sources

Name
Code
⊕ ChangeStatusSUP
Comments
⊕ CompanyIndSUP
Created By
Created On
DS1S
DS2S
DS3S
FlagSysSupPublish
⊕ IsActive
Last Updated By
Last Updated On
LastChangeUserDtm

Operators

IF AND
NOT OR
< >
<= >=
<> =
()
+ -

Functions

CHANGED
CHANGEDFROM
CHANGEDTO
CONCAT
CONTAINS
CONTAINSPATTERN
DATE
DATEADD
DATEDIFF
DATETIMEUTC

Expression

```
[SnOPSupplyProduct] = IF(INDEXOF(IF(LENGTH([SnOPDemandProduct].[MarketingCdNm] ) > 0 ,[SnOPDemandProduct].[MarketingCdNm] ,"" ) & IF(LENGTH([SnOPDemandProduct].[VariantIteration] ) > 0 , " " & [SnOPDemandProduct].[VariantIteration] ,"" ) & IF(IF(LENGTH([SnOPDemandProduct].[SnOPMktSwimlaneGrp]) > 0 , [SnOPDemandProduct].[SnOPMktSwimlaneGrp],"" ) = IF(LENGTH([SnOPDemandProduct].[SnOPMktSwimlane]) > 0 , [SnOPDemandProduct].[SnOPMktSwimlane],""), IF(LENGTH([SnOPDemandProduct].[SnOPMktSwimlaneGrp]) > 0 , " " & [SnOPDemandProduct].[SnOPMktSwimlaneGrp],""),IF(LENGTH([SnOPDemandProduct].[SnOPMktSwimlaneGrp]) > 0 , " " & [SnOPDemandProduct].[SnOPMktSwimlaneGrp],""),IF(LENGTH([SnOPDemandProduct].[SnOPMktSwimlaneGrp]) > 0 , " " & [SnOPDemandProduct].[SnOPMktSwimlaneGrp],""))
```

Display Description

[SnOPSupplyProduct] = IF(INDEXOF(IF(LENGTH([SnOPDemandProduct].[MarketingCdNm]) > 0 ,[SnOPDemandProduct].[MarketingCdNm] ,"") & IF(LE...

Save

Close

hmm.. my
eyes

What is
the
problem?

Problems



**Is this Logic?
Or is it a Liability?**

Display attribute

Complex logic

Too many little symbols

Trailing))



The "Syntax Tax"

- **Repetition:** The same logic is typed 3 times manually.
- **Fragility:** One missing comma breaks the whole string.
 - Trailing brackets
 - Commas
- **Opaque:** You can't "read" what the rule actually does at a glance.
 - Too many small symbols
 - Display attributes by square brackets
 - Complex chaining functions

THE PLANS

Nice to have search bar here as well

The screenshot shows the 'Expression Editor for Valid If' interface. It features three main panels: 'Data Sources', 'Operators', and 'Functions'. The 'Data Sources' panel lists attributes like Name, Code, Date1, and Date2, with a search bar at the top. The 'Operators' panel shows logical and comparison operators. The 'Functions' panel lists various date and string functions, with 'DATEADD' highlighted. The 'Expression' panel displays the current expression: `[Date2] <> DATEADD( [Date1] , 10 , "Month" )`. Below the expression, a visual representation of the expression is shown with tokens: `Date2`, `<>`, `DATEADD`, `Date1`, `10`, and `Month`. Annotations with red arrows point to specific features: a search bar in the Data Sources panel; a tooltip for the `DATEADD` function; a dropdown menu for the `Month` token; and a search dropdown for attribute names. A small icon in the bottom left corner consists of four dots in a 2x2 grid.

Expression Editor for Valid If

Data Sources

- Name
- Code
- Created By
- Created On
- Date1
- Date2
- Date3
- DBACustomer1
- DBACustomer2
- DBACustomer3
- DBACustomer4
- DBARecursive
- Last Updated By
- Last Updated On
- Link

Operators

- NOT
- OR
- <
- >
- <=
- >=
- <>
- =
- (
-)
- +
-
- *
- /

Functions

- CONCAT
- CONTAINS
- CONTAINSPATTERN
- DATE
- DATEADD**
- DATEDIFF
- DATETIMEUTC
- ENDSWITH
- IN
- INDEXOF

Expression

`[Date2] <> DATEADD( [Date1] , 10 , "Month" )`

Visual representation of the expression:

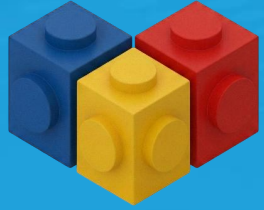
- `Date2`
- `<>`
- `DATEADD`
- `Date1`
- `10`
- `Month`

Annotations:

- Search for attribute name
- DATEADD: Adds the corresponding day/month/year to a given Date expression
- Month: Drop down with 3 options (Day, Month, Year)
- Drop down to search & change attribute quickly
- Quick details about Element on hover
- Only take Data Source type
- Only take number type

Buttons: Verify, Cancel

But where is the “flow”?



Proudly introduce

expression flow.

see the logic. skip the syntax.

•• Profisee

Team Thuan and 99+ AI Agents



Typing & Praying" → "Dragging & Snapping.

Logic, Visualized.

Valid If

Returns: Boolean (true / false)

⋮

▼

=

Equal to

×

LEFT

Name

RIGHT

⋮

▼

{ }

CONCAT

Concatenate two or more strings

×

VALUE 1

Code

VALUE 2

Color Name

VALUE 3

ProductLine Code

VALUE 4

ProductSubcategory Code

VALUE 5

Size Name

+ Add value

Change to

Returns: Value (string, number, etc.)

↔

⋮

▼

{ }

IF

Conditional: if/then/else

×

CONDITION

⋮

▼

>

Greater than

×

LEFT

⋮

▼

{ }

TODAY

Today's date

×

RIGHT

SellEndDate

TRUE CASE

T "CLEARANCE"

FALSE CASE

⊗ NULL

Zero. Syntax. Errors

Valid If

Returns: Boolean (true / false)

IN

Value is in the list

ATTRIBUTE

T "ABC"

VALUES

{ } ()

Parentheses grouping / value list

VALUE 1

Name

VALUE 2

DocumentationURL

VALUE 3

ProductSubcategory
Name

+ Add value



- "Impossible to forget a closing bracket."
- "The system handles the commas. You handle the strategy."

```
</> GENERATED  
Valid If bool  
"ABC" IN ([Name], [DocumentationURL],  
[ProductSubcategory].[Name])  
When bool  
/* empty */
```

Validation & Assignment (before)

Validation (0)

Assignments (0)

Validation Clauses

Valid If	When	Constraint
Click to create expression...	Click to create expression...	☐
Click to create expression...	Click to create expression...	☐
Click to create expression...	Click to create expression...	☐

Validation & Assignment (after)

**Valid If**
Returns: Boolean (true / false)

+

Drop an expression here
Start with a comparison operator (=, >, <, ...) or a logical function (AND, OR, ...)
e.g. [Code] = CONCAT([Class].[Name], [Color].[Name])

Paste Expression

**When**
Returns: Boolean (true / false)

+

Drop an expression here
Start with a comparison operator (=, >, <, ...) or a logical function (AND, OR, ...)
e.g. [Code] = CONCAT([Class].[Name], [Color].[Name])

Paste Expression

**Change to**
Returns: Value (string, number, etc.)

+

Drop an expression here
Start with a function (IF, CONCAT, ...) or a data attribute
e.g. IF(LENGTH([Name]) > 0, [Name], '')

Paste Expression

**When**
Returns: Boolean (true / false)

+

Drop an expression here
Start with a comparison operator (=, >, <, ...) or a logical function (AND, OR, ...)
e.g. [Code] = CONCAT([Class].[Name], [Color].[Name])

Paste Expression

Functions & Operators (before)

Operators		Functions
IF	AND	CHANGED
NOT	OR	CHANGEDFROM
<	>	CHANGEDTO
<=	>=	CONCAT
<>	=	CONTAINS
(	)	CONTAINSPATTERN
+	-	DATE
		DATEADD
		DATEDIFF
		DATETIMEUTC

Functions & Operators (after)

Search functions...

LOGIC

4

STRING

10

DATE & TIME

7

CHANGE DETECTION

4

CONVERSION

5

SET OPERATIONS

1

COMPARISON

6

ARITHMETIC

4

GROUPING

1

LOGIC

4

IF

3

Conditional: if/then/else

AND

2

Logical AND

OR

2

Logical OR

NOT

1

Logical NOT

ARITHMETIC

4

+

2

Addition

-

2

Subtraction

*

2

Multiplication

/

2

Division

COMPARISON

6

>

2

Greater than

<

2

Less than

>=

2

Greater or equal

<=

2

Less or equal

<>

2

Not equal to

=

2

Equal to

STRING

10

CONCAT

2

Concatenate two or more strings

CONTAINS

2

Checks if string contains substring

CONTAINSPATTERN

2

Checks if string contains pattern

ENDSWITH

2

Checks if string ends with value

STARTSWITH

2

Checks if string starts with value

INDEXOF

2

Finds index of string within string

LEFT

2

Returns N chars from the left

RIGHT

2

Returns N chars from the right

LENGTH

1

Returns string length

SUBSTRING

3

Returns a subset of a string

DATE & TIME

7

DATE

3

Returns a Date from DD/MM/YYYY

DATEADD

3

Add day/month/year to a date

DATEDIFF

2

Days between two dates

DATETIMEUTC

7

Returns a DateTime from components

RELATIVEDATE

2

Relative date (StartOfMonth, etc.)

NOW

const

Current date and time

TODAY

const

Today's date

And more...

Blocks & Custom Values & Types & Instant Edit

{} CONCAT

Concatenate two or more strings

VALUE 1

Class Name

VALUE 2

Color Name

VALUE 3

+ Drop Value 3 here

+ Add value

+ CUSTOM VALUE

This is custom value

T "This is custom value" drag me Text

EMPTY SPACE NULL TRUE FALSE 0 1

VALUE 1

T "ABC"

VALUE 2

999

VALUE 3

TRUE

VALUE 4

NULL

T

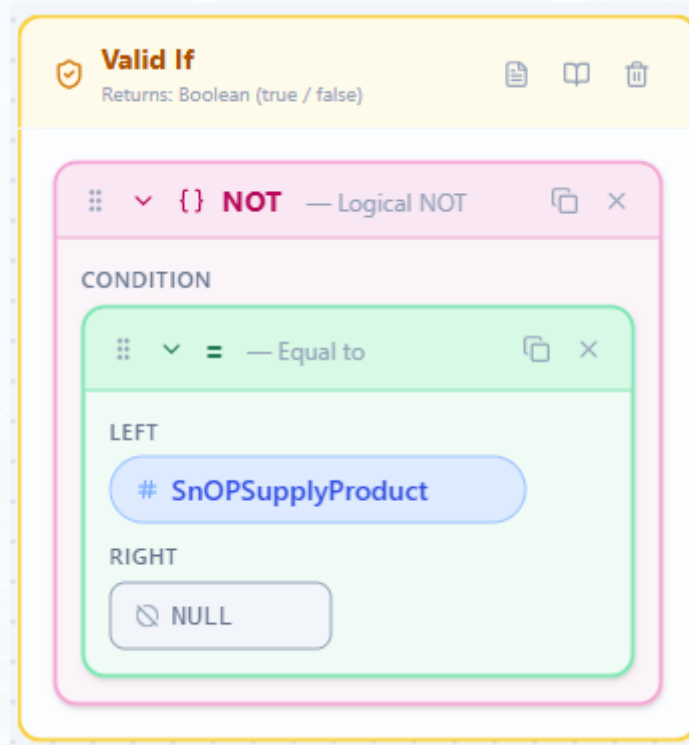
Profisee

Text Expression (before)

Expression

NOT([SnOPSupplyProduct] = NULL)

Block Expression (after)



A few descriptions (before)

The screenshot displays a software interface with three main components:

- Data Panel (Left):** A list of data fields with expandable/collapsible icons (minus signs). The visible fields are:
 - BaseUOM
 - ChangeStatusD
 - Name
 - Code
 - Comments
 - CompanyIndDM
 - Name
 - Code
 - CompanyIn
 - DependentDBA
 - DesignBizCd
 - Name
 - Code
 - DivisionCd
 - DivisionDsc
- CHANGED Function Description (Center Popup):** A light blue box with a close button (X) in the top right corner. It contains the following text:
 - CHANGED** (with an information icon 'i')
 - Checks for any value change in the specified list of attributes. These cannot be Multi-level Attributes. Only used as a conditionExpression in an IF statement.
 - Syntax:**
CHANGED(<attribute1> , <attribute2> , ..., <attributeN>)
 - Parameters:**
attribute1: An attribute
Allowed Values: Attribute
 - attribute2: An attribute
Allowed Values: Attribute
- Functions Panel (Right):** A list of functions with a scroll bar on the right. The functions listed are:
 - CHANGED (with an information icon 'i')
 - CHANGEDFROM
 - CHANGEDTO
 - CONCAT
 - CONTAINS
 - CONTAINSPATTERN
 - DATE
 - DATEADD
 - DATEDIFF
 - DATETIMEUTC

Description for all (after)

The screenshot displays a logic builder interface. On the left, a sidebar lists various logic categories: LOGIC, STRING, DATE & TIME, CHANGE DETECTION, CONVERSION, SET OPERATIONS, COMPARISON, ARITHMETIC, and GROUPING. The 'LOGIC' category is expanded, showing a list of functions: IF (Conditional: if/then/else), AND (Logical AND), OR (Logical OR), and NOT (Logical NOT). A tooltip for the 'IF' function is open, providing its syntax and parameters.

IF
Conditional: if/then/else

Syntax: IF(<conditionExpression>, <trueExpression>, <falseExpression>)

Parameters:

- conditionExpression – Boolean expression to evaluate
- trueExpression – Returned when condition is true
- falseExpression – Returned when condition is false

Data sources

Data Sources

- Name
- Code
- ⊕ Class
- ⊕ Color
- Created By
- Created On
- DealerCost
- DocumentationURL
- ⊕ InHouseManufactured
- Last Updated By
- Last Updated On
- MSRP
- ⊕ ProductLine
- ⊕ ProductSubcategory
- ReorderPoint



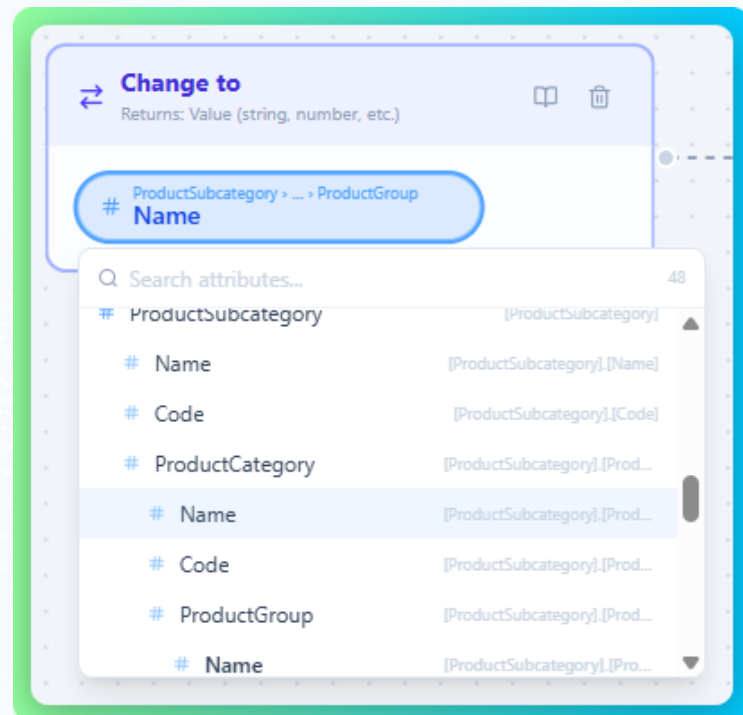
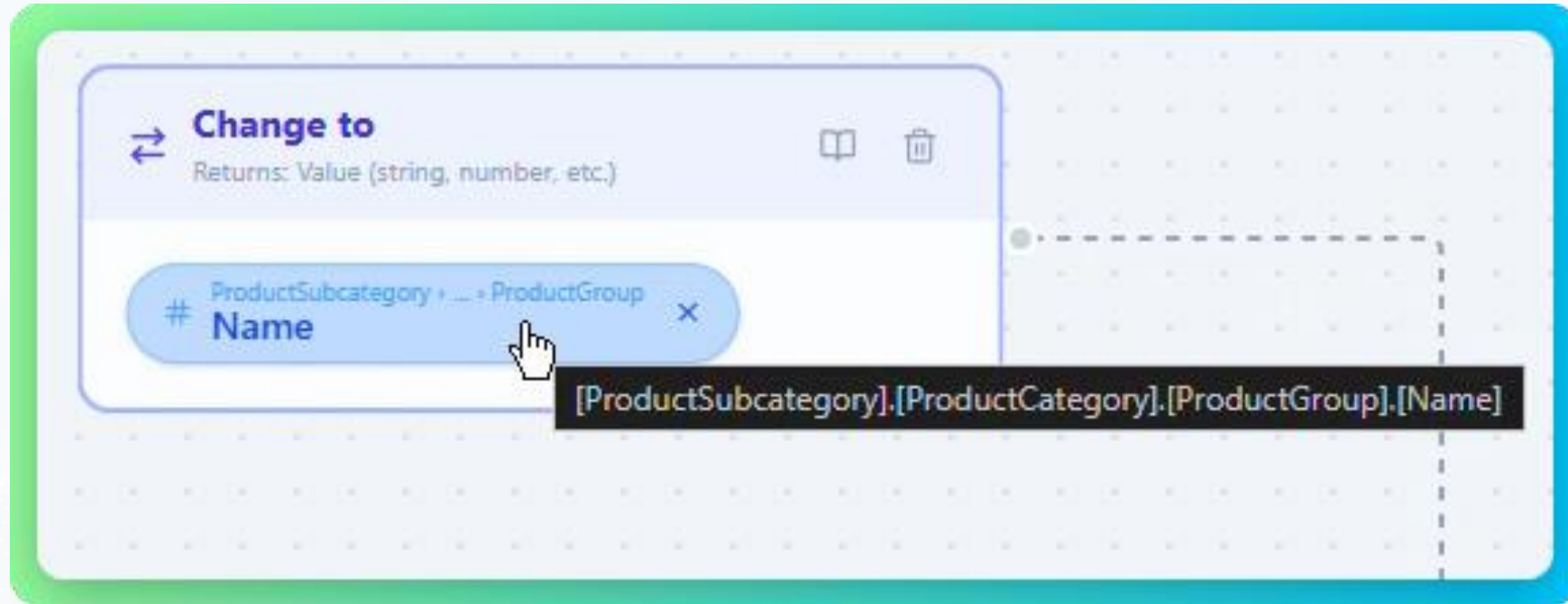
Product

24

ATTRIBUTES

- ProductLine +2
 - Name
 - Code
- ProductSubcategory +3
 - Name
 - Code
- ProductCategory +3
 - Name
 - Code
- ProductGroup +2
 - Name
 - Code

Attribute Nodes Quick Lookup & Replacement



Drag-N-Drop

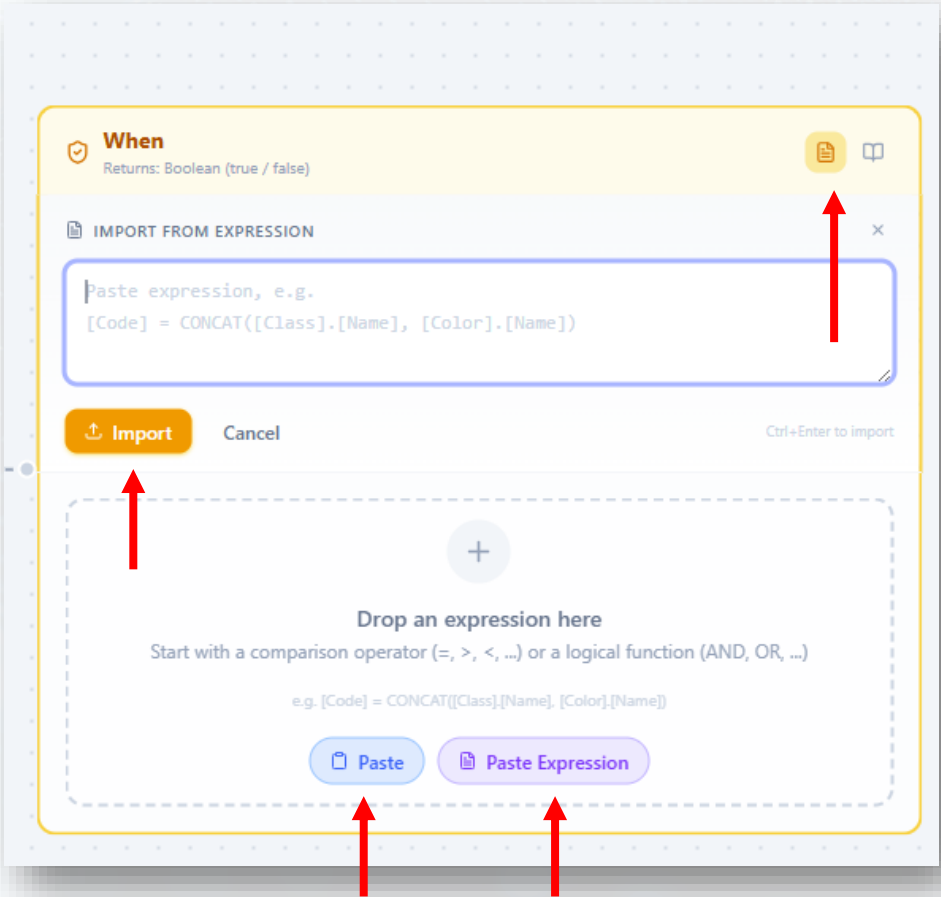
The screenshot shows a 'Change to' block with a light purple header and a dashed blue box for an expression. A 'DATEADD' block is being dragged from a palette on the left into the dashed box. The 'DATEADD' block has a green header, the text 'Add day/month/year to a date', and a value of '3'. The 'Change to' block has a double-headed arrow icon, the text 'Change to', and 'Returns: Value (string, number, etc.)'. The dashed box contains the text 'Drop an expression here', 'Start with a function (IF, CONCAT, ...) or a data attribute', and an example 'e.g. IF(LENGTH([Name]) > 0, [Name], '')'.

The screenshot shows a 'Valid If' block with a light orange header and a blue body. The header contains a shield icon, the text 'Valid If', and 'Returns: Boolean (true / false)'. The body contains a 'CONCAT' block with a blue header and the text 'Concatenate two or more strings'. The 'CONCAT' block has four input fields labeled 'VALUE 1', 'VALUE 2', 'VALUE 3', and 'VALUE 4'. The first three fields contain '# Name', '# Code', and '# MSRP' respectively. The fourth field contains '# SellStartDate'. A dashed blue box is overlaid on the 'VALUE 4' field with the text 'Drop Value 4 here'. Below the input fields is a button labeled '+ Add value'.

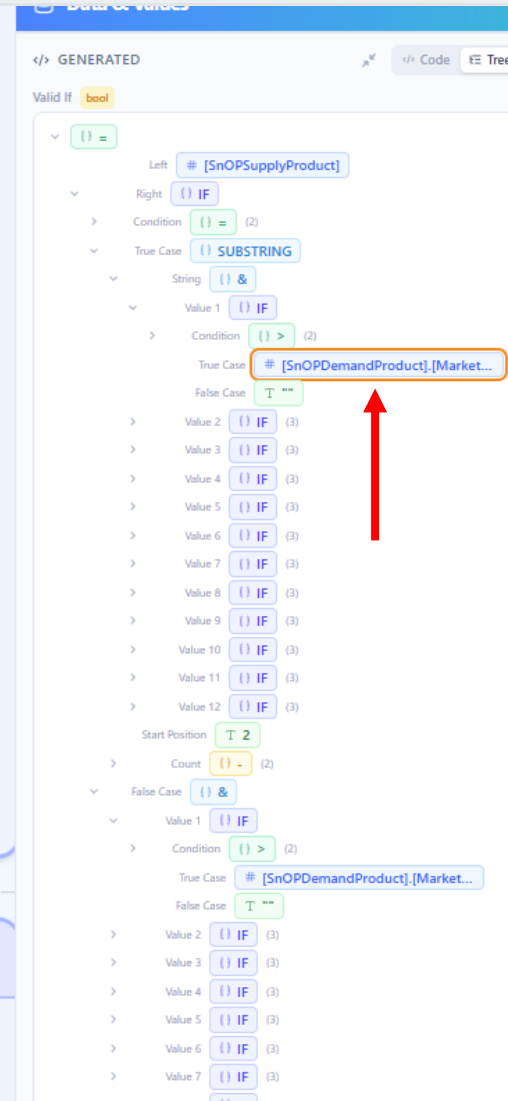
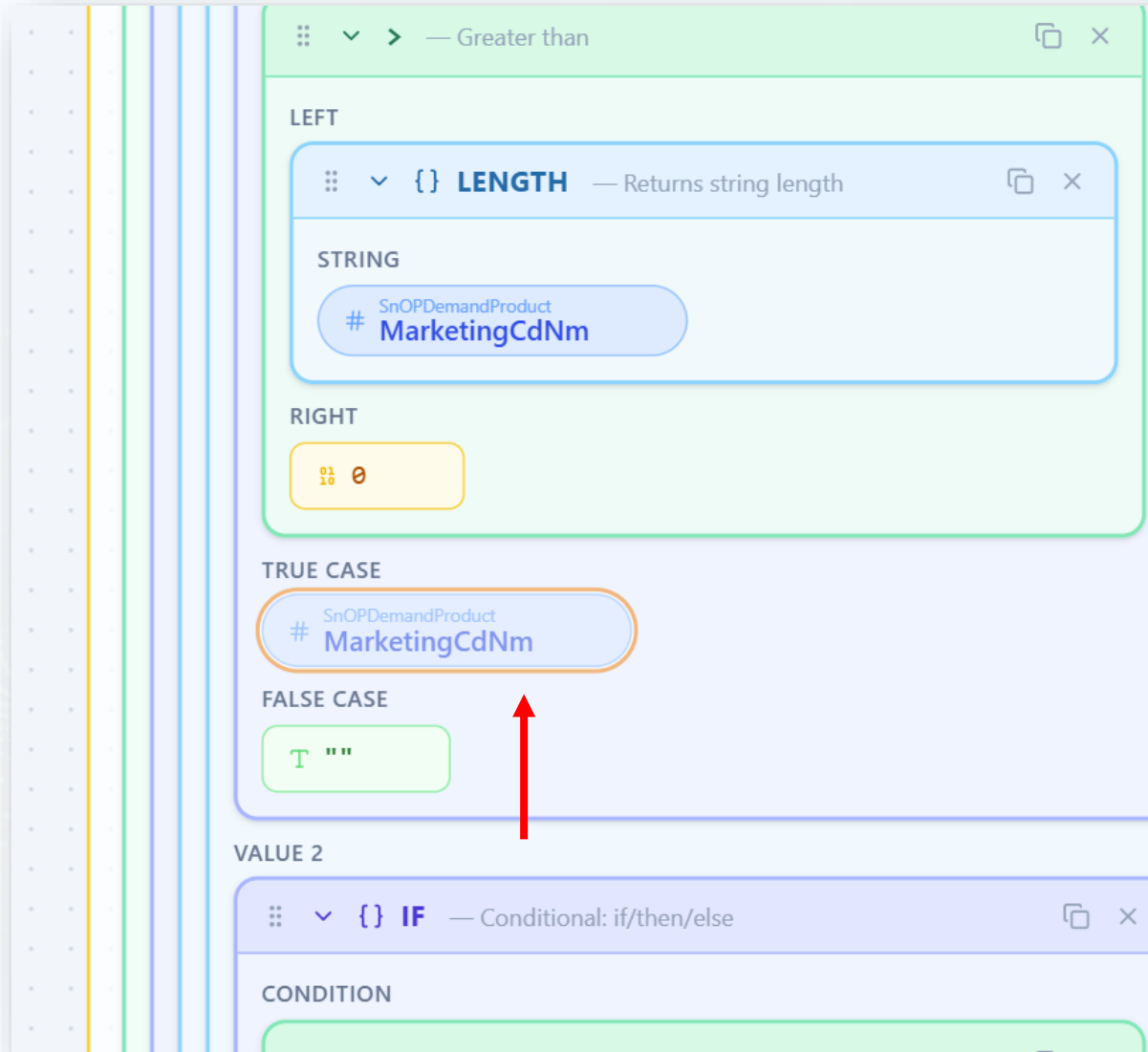
Copy-N-Paste

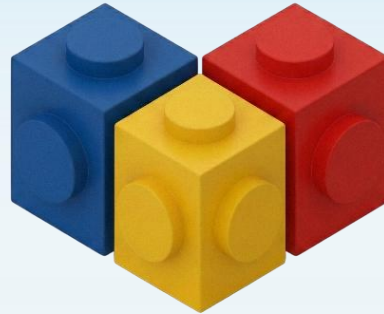


Convert Text Expression to Blocks



Autofocus Tree view for Impossible Expression





Expression Flow

Stop Writing Code. Start Building Rules.